RUI XING

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TECHNICAL SKILLS

Core Domain	Large-Scale Optimization, GenAI Agents, and AWS ML Infrastructure.
Language	Java, Python, C, C++, Typescript, SQL, Bash
Technologies	AWS (Bedrock, ECS, Lambda, Stepfunction, DynamoDB), CDK, Docker

RELEVANT WORK EXPERIENCE

Amazon Bellevue, WA
Software Development Engineer **2020 – Present**

Large-Scale Order Quantity Optimization Engine **2025**

- Architected and engineered an end-to-end scalable system for large-scale Order Quantity Optimization for Amazon's sourcing team, utilizing a novel Column Generation decomposition framework to manage million-variable resource models, improved order quantity optimization efficiency.
- Developed the core high-performance engine in Java, implementing custom data structures and a selective variable pricing mechanism that minimized memory overhead and reduced model convergence runtime by 9.5%.
- Optimized system throughput by implementing multi-threaded execution for candidate evaluation, achieving a 20% reduction in latency and maximizing hardware utilization during high-load scenarios.
- Designed a robust constraint verification framework to ensure algorithmic fidelity; integrated ML-based heuristics (Linear Regression/Decision Trees) to dynamically predict and optimize execution paths.

Complex Workflow LLM Agent/GenAI Observability **2025**

- Initiated and built a GenAI MVP to monitor massive-scale distributed discrete-event simulations, benchmarking near-real-time run velocity against historical baselines to deliver one-click anomaly detection for managers.
- Implemented end-to-end workflow using AWS Bedrock agentic orchestration and Lambda-backed APIs to query heterogeneous sources, generate evidence-linked investigation narratives, cutting manual drill and on-call toil.
- Demoed the MVP to the team and drove alignment on the reference workflow; catalyzed a cross-team GenAI observability direction and enabled a sister team to take ownership for productionization.

Simulation Fidelity & Sensitivity Analysis **2024**

- Developed a sensitivity-analysis framework to de-risk an upstream dependency deprecation, running high-concurrency dataset perturbations and parallel service calls to quantify data-fidelity gaps and guide rollout.

Data Modeling **2023**

- Re-architected a production DynamoDB data model with an access-pattern-driven key design to preserve historical state, eliminate update-driven data loss, and improve auditability.
- Designed a backward-compatible, zero-downtime migration using an idempotent one-time backfill plus stream-driven change propagation for live updates.

MLE Execution Platform **2022**

- Built the 0→1 AWS IaC (Infrastructure as Code) foundation for a production optimization service, standardizing multi-environment deployments and release governance through a reproducible framework.
- Engineered a greenfield event-driven orchestration engine (EventBridge, Lambda, ECS) to execute containerized scientist-authored Python workflows, enabling parallel task runs that reduced end-to-end runtime by 33%.

EDUCATION

Bachelor of Computer Science, Honours, Co-op **January 2016 - June 2020**
University of Waterloo, Canada